



12-Week OpenCV Course: From Face Detection to Reliable Recognition

Week 1 — The Big Picture

Goal: Understand what the program is doing before worrying about individual lines.

- What happens from the moment the webcam sees a person until the computer speaks their name?
- Draw the complete pipeline.
- What parts of the program are OpenCV?
- What parts are Python?
- What parts are third-party libraries?
- What part is responsible for the camera?
- What part detects a face?
- What part recognizes the person?
- What part generates speech?
- What part plays the audio?

Question:

Is face detection the same thing as face recognition?

Week 2 — Images Are Data

Goal: Understand what OpenCV actually receives from the camera.

- What is a pixel?
 - What is an image array?
 - What is image resolution?
 - What does `frame` contain?
 - What is `ret` ?
 - Why does `camera.read()` return two values?
 - What is BGR?
 - Why does OpenCV traditionally use BGR instead of RGB?
 - What happens when an image is converted to grayscale?
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- Why might grayscale be useful?

Experiment:

- Change the webcam resolution.
 - Save frames using `cv2.imwrite()`.
 - Open the saved images.
 - Compare low and high resolution.
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Week 3 — Webcam and Real-Time Processing

Goal: Understand why the program needs a continuous loop.

Investigate:

- What does `cv2.VideoCapture(0)` do?
- What does the `0` mean?
- What is a frame?
- Why do we need a `while` loop?
- What does `cv2.imshow()` do?
- What does `cv2.waitKey()` do?
- Why does the program need `camera.release()` ?
- What happens if we remove `destroyAllWindows()` ?

Research question:

What is FPS, and why does it matter for computer vision?

Experiment:

How does your program behave at 15 FPS, 30 FPS, and 60 FPS?

Week 4 — Face Detection

Goal: Understand how OpenCV finds a face.

Introduce:

```
1 | haarcascade_frontalface_default.xml
2 |
```

- What is a Haar Cascade?
 - Who developed the Haar Cascade approach?
-

- Is it machine learning?
- Is it deep learning?
- What does the XML file contain?
- Why does OpenCV need this file?
- What does `detectMultiScale()` do?
- What does `scaleFactor` mean?
- What does `minNeighbors` mean?
- What does `minSize` mean?

Experiment:

Try:

```
1 minNeighbors = 3
2 minNeighbors = 5
3 minNeighbors = 8
4 minNeighbors = 15
5
```

Ask:

Which setting gives more false detections?

Week 5 — Detection Is Not Recognition

Goal: Understand the fundamental distinction.

- If OpenCV draws a box around your face, does it know your name?
- What is face detection?
- What is face recognition?
- Why are they two separate problems?
- Could a system detect a face but not recognize the person?
- Could recognition work if detection fails?

Draw:

```
1 Camera
2   ↓
3 Face Detection
4   ↓
5 Face Recognition
6   ↓
7 Name
8
```

Then ask:

Where could the system fail?

Week 6 — Training Data

Goal: Understand where the "knowledge" comes from.

Given:

```
1 faces/  
2 |— Karthik/  
3 |— Lynette/  
4 |— Ethan/  
5
```

- Why are the images organized into folders?
- Why is the folder name useful?
- What is a label?
- Why does the computer need numerical labels?
- How does `Karthik` become a number?
- Where are the labels stored?
- What happens if you add another person?
- Do you need to retrain?
- Where is the trained model stored?

Then investigate:

```
1 faces.yml  
2 names.txt  
3
```

Experiment:

Train with:

- 1 image
- 5 images
- 10 images
- 20 images

Compare recognition.

Week 7 — LBPH

Goal: Understand the actual recognition algorithm.

Research:

```
1 cv2.face.LBPHFaceRecognizer_create()  
2
```

Ask:

- What does LBPH stand for?
- What problem does LBPH solve?
- What information does LBPH use from a face?
- Why does LBPH use local patterns?
- What is a histogram?
- What does "local binary pattern" mean?
- What are the strengths of LBPH?
- What are its weaknesses?
- Is LBPH a deep-learning model?

Then investigate:

```
1 label, distance = recognizer.predict(face)  
2
```

- What is `label` ?
 - What is `distance` ?
 - Is a lower distance value better or worse?
 - Why?
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Week 8 — Make It Fail

Goal: Students learn that computer vision systems are imperfect.

challenge:

Find as many ways as possible to make the recognition fail.

Test:

- Bright light
 - Dim light
 - Darkness
 - Face looking left
 - Face looking right
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- Face looking up/down
- Glasses
- Hat
- Different facial expressions
- Face very close to camera
- Face far from camera
- Fast movement
- Multiple people

For every failure, ask:

What failed?

Possible answers:

- Camera
- Image quality
- Face detection
- Training data
- LBPH recognition
- Confidence threshold

Week 9 — Hardware vs Software

Goal: Understand that computer vision starts with physical sensors.

Ask:

- Why doesn't the webcam work well in darkness?
 - What happens to the image in low light?
 - What is sensor noise?
 - What is exposure?
 - What is gain?
 - What is motion blur?
 - What is dynamic range?
 - Can software recover information the camera never captured?
 - What makes a camera good for low-light vision?
 - What is an infrared camera?
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- Why do some security cameras work in darkness?

Experiment:

Run the same program in:

- Bright room
- Normal room
- Dim room
- Very dark room

Then ask:

Did you change the OpenCV algorithm?

If not:

Why did the result change?

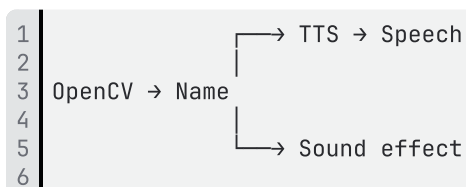
Week 10 — From Vision to Voice

Goal: Understand that the computer-vision system and audio system are separate.

Investigate:

- What is text-to-speech?
- What is `edge-tts` ?
- What is a neural voice?
- What is an MP3?
- What is the difference between generated speech and prerecorded audio?
- Why does `afplay` play the MP3?
- Why does playing audio sometimes freeze the camera?
- What does "blocking" mean?
- Why do we use a background thread?
- What is `threading.Thread()` doing?

Draw:



Week 11 — Making the System Reliable

Goal: Move from "it works" to "it works reliably."

problems:

- Why does it sometimes call Karthik "Unknown"?
- Why does it detect objects as unknown faces?
- Why does the name change from Karthik → Unknown → Karthik?
- Why does it sometimes announce repeatedly?
- Why do we need a 5-second timer?
- Why might we require several consecutive frames?
- Why do we need an `audio_playing` flag?
- What happens if two audio announcements start simultaneously?

Task:

How would you make the system more stable without replacing the entire algorithm?

Week 12 — How Would a Professional System Work?

Goal: Connect the classroom project to modern computer vision.

Given the existing system:

```
1 Haar Cascade
2   ↓
3 LBPH
4
```

Research:

- Why are Haar Cascades considered an older approach?
 - Why does modern computer vision use deep learning?
 - What is a CNN?
 - What is a face embedding?
 - How does an embedding represent a face?
 - What is similarity/distance?
 - How is a modern face detector different from Haar?
 - What is face anti-spoofing?
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- What is liveness detection?
- Why might a photograph fool a simple face-recognition system?
- Would you use LBPH in a commercial product? Why or why not?

Final presentation

should be able to explain:

1. **How the camera produces an image**
 2. **How OpenCV finds a face**
 3. **How LBPH recognizes a person**
 4. **How training data affects recognition**
 5. **Why the system fails**
 6. **How hardware affects computer vision**
 7. **How the audio system works**
 8. **Why threading is necessary**
 9. **How you would improve the system**
 10. **How modern systems differ from this project**
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